

CHICAGO HARRISON-DEVER CRIB, IL, USA

WMO: 997338

Lat: 41.916N

Lon: 87.572W

Elev: 202

StdP: 98.92

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -16.6	(c) -13.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 18.9	(k) -4.6	(l) 17.8	(m) -3.1	(n) 8.8	(o) 300	(p) 0.747

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 5.1	(c) 30.6	(d) N/A	(e) 29.0	(f) N/A	(g) 27.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 8.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.5	(b) 15.0	(c) 13.4	(e) DB	(f) -19.6	(g) 32.5	(h) 4.2	(i) 2.2	(j) -22.6	(k) 34.0	(l) -25.0	(m) 35.3	(n) -27.4	(o) 36.5	(p) -30.4	(q) 38.1	(r) N/A
(4) 16.5	(5) 15.0	(6) 13.4	(7) DB	(8) -19.6	(9) 32.5	(10) 4.2	(11) 2.2	(12) -22.6	(13) 34.0	(14) -25.0	(15) 35.3	(16) -27.4	(17) 36.5	(18) -30.4	(19) 38.1	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-3.9	-3.2	2.8	8.2	14.0	19.5	22.8	22.5	19.1	12.5	6.0	-0.4		
	DBStd	10.48	5.86	5.63	5.53	4.37	4.82	4.22	3.30	2.48	3.60	4.42	5.36	5.14		
	HDD10.0	1625	432	371	232	87	15	0	0	0	0	23	141	324		
	HDD18.3	3452	690	604	482	306	153	38	4	2	32	188	371	582		
	CDD10.0	1640	0	1	9	33	138	284	396	387	272	101	19	1		
	CDD18.3	425	0	0	1	1	18	71	142	131	54	7	0	0		
	CDH23.3	2107	0	0	1	8	109	396	822	517	227	28	0	0		
	CDH26.7	492	0	0	0	0	18	83	236	99	53	3	0	0		
(14) Wind	WSAvg	7.1	8.1	7.7	7.3	7.5	6.7	6.1	5.6	5.5	6.5	7.8	8.3	7.7		
(15) Precipitation	PrecAvg	906	44	37	64	92	87	92	95	111	88	67	70	59		
	PrecMax	1253	113	112	150	195	186	253	212	434	360	217	209	217		
	PrecMin	566	2	5	16	19	8	24	28	17	0	5	12	8		
	PrecStd	162	29	23	31	42	46	50	43	83	69	45	46	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.0	22.2	24.7	29.1	31.7	34.2	31.1	30.7	26.4	19.7	14.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	9.3	10.1	18.9	21.5	26.4	28.9	31.3	29.5	28.0	23.4	17.2	10.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	6.5	6.3	15.2	18.9	24.3	27.2	29.4	27.6	25.8	21.1	15.1	8.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	3.9	3.5	10.7	15.7	21.8	25.6	27.6	25.9	24.0	18.8	13.0	6.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	5.8	6.0	6.4	7.0	6.3	5.1	4.4	5.0	5.5	5.7	5.3		
		MCDBR	8.2	9.5	12.3	12.8	10.6	9.0	8.2	7.0	7.7	7.8	7.9	7.6		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.322	0.348	0.403	0.432	0.440	0.448	0.445	0.416	0.361	0.321	0.304		
	taud		2.461	2.388	2.381	2.262	2.238	2.267	2.243	2.251	2.296	2.450	2.515	2.504		
	Ebn at Noon		848	882	897	866	847	839	826	817	815	828	821	817		
	Edh at Noon		75	95	107	130	137	133	135	130	116	89	72	67		
(44) All-Sky Solar Radiation	RadAvg		1.55	2.40	3.56	4.61	5.25	6.04	6.11	5.30	4.36	2.86	1.80	1.29		
	RadStd		0.17	0.31	0.27	0.38	0.44	0.39	0.28	0.31	0.37	0.31	0.19	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page